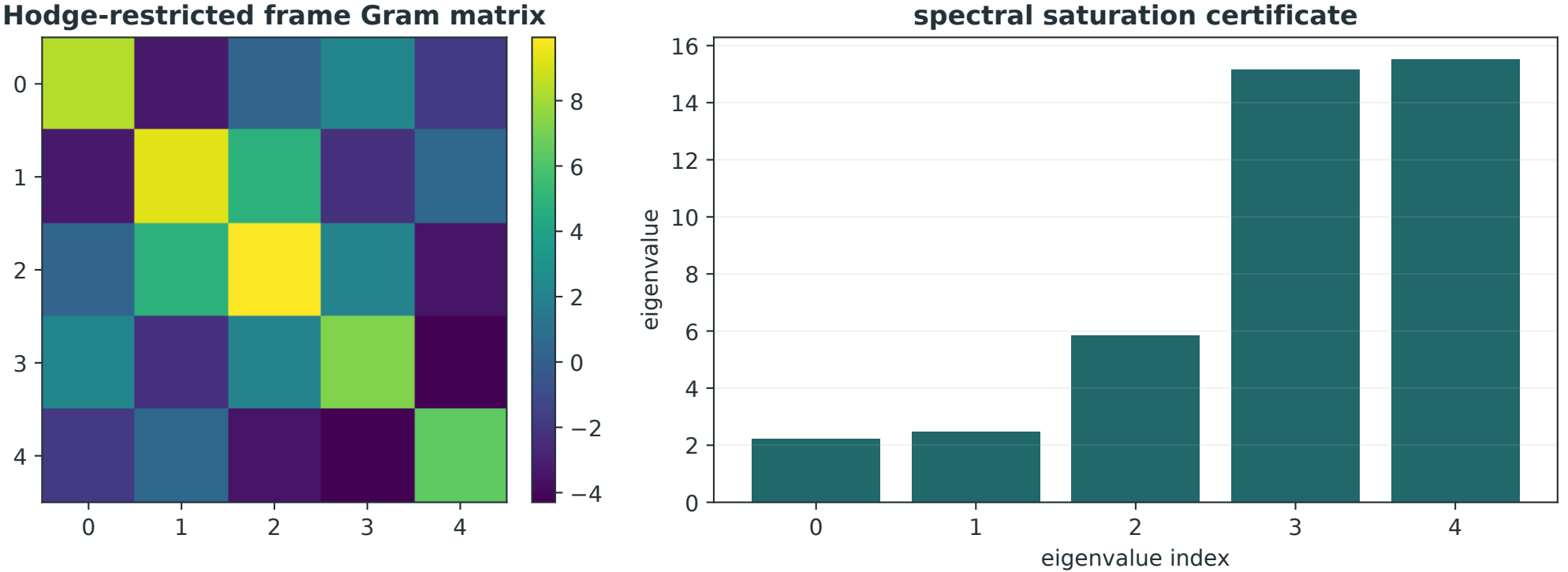A positive minimum eigenvalue certifies complete cycle coverage